

PREDICTING MODULE LATTICE REDUCTION

Code link: <https://github.com/lucas/mBKZ>

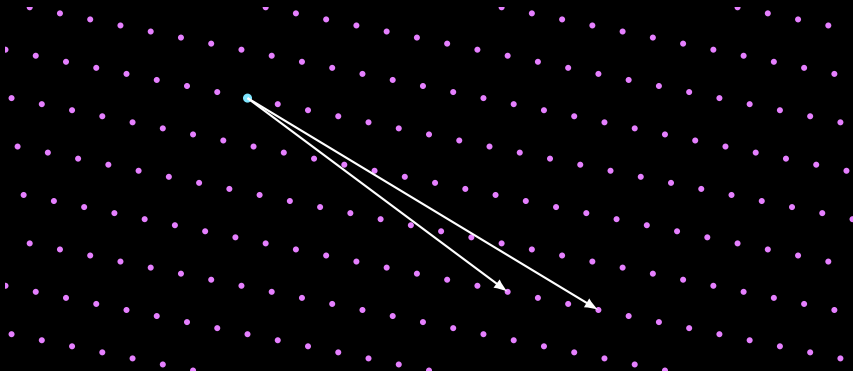
Archive Link: <https://github.com/lucas/mBKZ/blob/main/mBKZ.pdf>

Paper by Léo Ducas^{1,2}, Lynn Engelberts^{1,3} and Paola de Perthuis¹.

¹: CWI, ²: Leiden University, ³: QuSoft

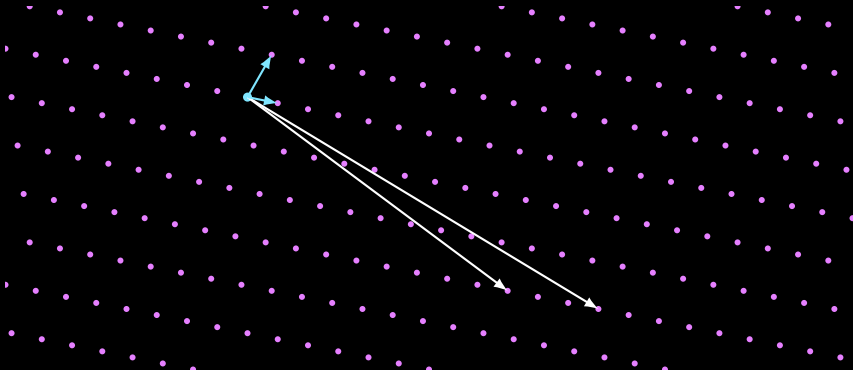
Introduction

Shortest Vector Problem



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Shortest Vector Problem



The BKZ Algorithm

Uses Gram-Schmidt orthogonalisation:

given basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_n$,
we denote: $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ the linear operation
projecting vectors orthogonally to:
 $\text{span}_{\mathbb{R}}\{\mathbf{b}_1, \dots, \mathbf{b}_{i-1}\}$.

The BKZ Algorithm

Algorithm 1: BKZ Algorithm Overview,
with an SVP oracle in dimension β

Input: LLL-reduced basis $\mathcal{B}$ of $\Lambda \subset \mathbb{R}^n$, blocksize β

```
1 while this step changes the basis  $\mathcal{B}$  do
2   for  $i$  in  $\llbracket 1; n \rrbracket$  do
3      $j \leftarrow \min(i + \beta - 1, n)$ 
4     Let  $\mathbf{v} \leftarrow \text{SVP} \left( \pi_i \left( \mathcal{B}_{\llbracket i; j \rrbracket} \right) \right)$ 
5     and  $\mathbf{w}$  such that  $\pi_i(\mathbf{w}) = \mathbf{v}$ 
6     Insert  $\mathbf{w}$  into  $\mathcal{B}$  at position  $i$ , remove linear
      dependencies using LLL
7 return  $\mathcal{B}$ 
```

Module Lattices

Number field $K = \mathbb{Q}[X]/P(X)$, where $P \in \mathbb{Q}[X]$ irreducible, of degree d .

$\mathcal{O}_K$ its ring of integers.

$$K_{\mathbb{R}} = K \otimes_{\mathbb{Q}} \mathbb{R}$$

I fractional ideal of $\mathcal{O}_K$: $\mathcal{O}_K$ -submodule of K s. t. $\exists x \in K^* \mid xI \subset \mathcal{O}_K$.

Module lattice M : defined with $\mathbf{b}_1, \dots, \mathbf{b}_r \in K_{\mathbb{R}}^{\rho}$ and $I_1, \dots, I_r$ fractional ideals of K , as $M = I_1 \mathbf{b}_1 + \dots + I_r \mathbf{b}_r$.

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Module lattice M : defined with $\mathbf{b}_1, \dots, \mathbf{b}_r \in K_{\mathbb{R}}^{\rho}$ and $I_1, \dots, I_r$ fractional ideals of K , as $M = I_1 \mathbf{b}_1 + \dots + I_r \mathbf{b}_r$.

One may obtain a lattice in $\mathbb{R}^{d\rho}$ from M , by embedding elements of $K_{\mathbb{R}}$ into $\mathbb{R}^d$.

Ex.: canonical embedding, coefficient embedding.

mBKZ: BKZ Generalized to Modules (stemming from [LPSW19]'s mLLL)

Algorithm 2: mBKZ Algorithm Overview,
with an SVP oracle in dimension β_K

Input: LLL-reduced unital pseudobasis $\mathcal{B} = (I_k, \mathbf{b}_k)_k$ of $M \subset K_{\mathbb{R}}^r$, blocksize β_K

```
1 while this step changes  $\mathcal{B}$  do
2   for  $i$  in  $\llbracket 1; r \rrbracket$  do
3      $j \leftarrow \min(i + \beta_K - 1, r)$ 
4      $\mathbf{v} \leftarrow \text{SVP}(\pi_i(\mathcal{B}_{\llbracket i; j \rrbracket}))$ 
5     Let  $I$  be such that  $\mathbf{v}I = \mathbf{v}K \cap \text{Module}(\pi_i(\mathcal{B}_{\llbracket i; j \rrbracket}))$ 
6     Let  $\mathbf{w}$  be such that  $\mathbf{w}I \subset M$  and  $\pi_i(\mathbf{w}) = \mathbf{v}$ 
7     Insert  $(\mathbf{w}, I)$  into  $\mathcal{B}$  at position  $i$ 
8   Remove linear dependencies
9 return  $\mathcal{B}$ 
```

How does mBKZ perform with respect to BKZ?

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Can we do better than embedding M in $\mathbb{R}^{rd}$ and applying BKZ?

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Context of Post-Quantum Standardizations

→ important to know whether the structure can be exploited

Overview of the Presentation

I. Introduction

II. Method: Generalizing Usual Heuristics to Module Lattices

A. The Geometric Series Assumption (GSA)

B. The Gaussian Heuristic

III. Our Analysis

A. Expressing the mBKZ slope

B. Analysis of the Terms in the mBKZ Slope

C. Analysis of Dimension Gains

IV. Conclusion

A. The Geometric Series Assumption (GSA)

1. IN THE UNSTRUCTURED CASE (introduced in [Sch03])

For $(\mathbf{b}_1, \dots, \mathbf{b}_n) \in (\mathbb{R}^n)^n$ a lattice basis, ordered with vectors of descending norm, we denote $(\mathbf{b}_1^*, \dots, \mathbf{b}_n^*)$ its Gram-Schmidt orthogonalisation, obtained recursively by setting:

$$\mathbf{b}_1^* = \mathbf{b}_1 \text{ and, for each } i \in \llbracket 2; n \rrbracket: \mathbf{b}_i^* = \pi_i(\mathbf{b}_i),$$

where $\pi_i^* : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the linear operation projecting vectors orthogonally to $\text{span}_{\mathbb{R}}\{\mathbf{b}_1^*, \dots, \mathbf{b}_{i-1}^*\}$.

GSA Heuristic

Using an algorithm $\mathcal{A}$ to reduce a basis, there is an $\alpha_{\mathcal{A}} \in [0; 1]$ such that:

$$\forall i \in \llbracket 1; n-1 \rrbracket, \|\mathbf{b}_{i+1}^*\| \approx \alpha_{\mathcal{A}} \|\mathbf{b}_i^*\|$$

A. The Geometric Series Assumption (GSA)

2. FOR MODULE BKZ

Norm and Determinant Notions:

Let $\sigma : K \rightarrow \mathbb{C}^d, x \mapsto (\sigma_1(x), \dots, \sigma_d(x))$ denote the canonical embedding from K to $\mathbb{C}$.

Algebraic norm on $K_{\mathbb{R}}$ $\mathcal{N} : K_{\mathbb{R}} \rightarrow \mathbb{R}, x \mapsto \prod_i |\sigma_i(x)|$,

algebraic norm of a fractional ideal I of $\mathcal{O}_K$: $\mathcal{N}(I) = \frac{|\mathcal{O}_K/xI|}{\mathcal{N}(x)}$ for any $x \in K^* \mid xI \subset \mathcal{O}_K$.

$\langle \cdot, \cdot \rangle : K_{\mathbb{R}}^2 \rightarrow \mathbb{C}, (x, y) \mapsto \sum_i \sigma_i(x) \overline{\sigma_i(y)}$ defines the *trace* norm $\|\cdot\| : K_{\mathbb{R}} \rightarrow \mathbb{R}, x \mapsto \sqrt{\langle x, x \rangle}$, extended to $K_{\mathbb{R}}^r$ in the natural way (for Gram-Schmidt orthogonalisation).

Given $(\mathbf{b}_k, I_k)$ a pseudobasis of M , $B = (\mathbf{b}_k)_k$, we define:

$$\det(M) = \sqrt{|\Delta_K|^r} \sqrt{\mathcal{N} \left(\det \left(\overline{B}^T B \right) \right)} \prod_k \mathcal{N}(I_k).$$

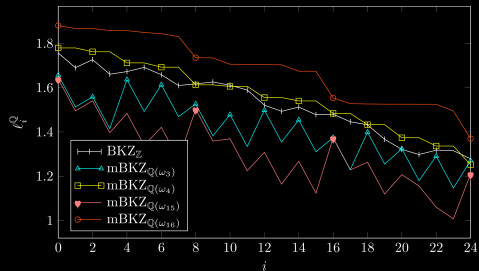
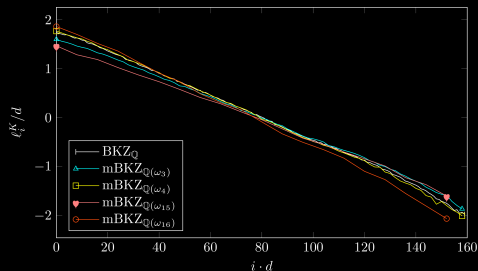
A. The Geometric Series Assumption (GSA)

2. FOR MODULE BKZ

We Adapt the GSA Heuristic

For $(\mathbf{b}_k, I_k)_k$ a pseudobasis of $M \subset K_{\mathbb{R}}^r$ and $(\mathbf{b}_k^*)_k$ its Gram-Schmidt projection:

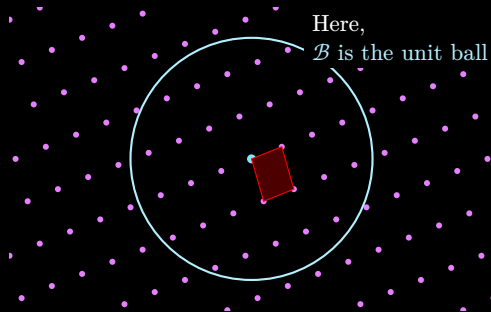
$$\forall k \in \llbracket 1; r-1 \rrbracket, \det(\mathbf{b}_k^* I_k) \approx \alpha \det(\mathbf{b}_{k-1}^* I_{k-1}).$$



B. The Gaussian Heuristic

1. IN THE UNSTRUCTURED CASE

An example:

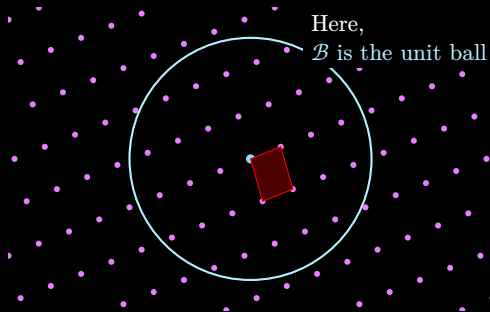


B. The Gaussian Heuristic

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An example:

31 points inside $\mathcal{B}$, and $\frac{\text{Vol}_{\mathcal{B}}}{\text{Vol}_{\Lambda}} \approx 32$.



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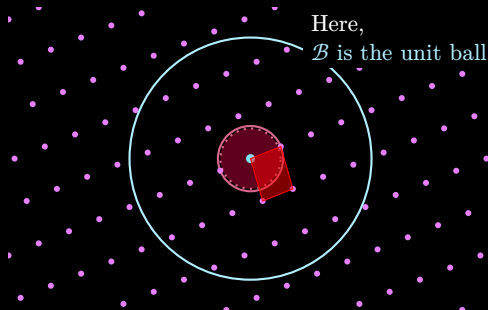
An example:

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And approximation of shortest vector length as radius of ball with two points:

$$\lambda_1 \approx \left(\frac{2\text{Vol}_{\Lambda}}{\text{Vol}_{\mathcal{B}}} \right)^{\frac{1}{d}},$$

where d is the dimension.



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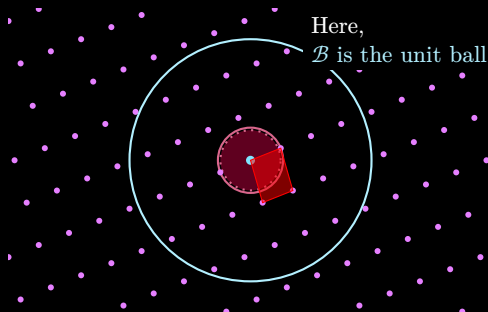
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where d is the dimension.

Here: $\lambda_1 = 0.27 \approx 0.25$.



B. The Gaussian Heuristic

2. A GAUSSIAN HEURISTIC FOR MODULES (FROM [GSVV24])

The number field K of degree d 's geometry yields new factor in λ_1 approximation for a module lattice $M \subset K_{\mathbb{R}}^r$.

Indeed, given a vector, other vectors of the same Euclidean norm will be provided by multiplication with roots of unity in K .

Hence the ball of radius λ_1 will hold $i \cdot \frac{\mu_K}{2} + 1$ points, for an integer i in $\mathbb{N}^*$ and μ_K the number of roots of unity in K . In our heuristics, we will consider that it will contain $\frac{\mu_K}{2}$ points.

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Expressing the Module-BKZ (mBKZ) Slope α_K

We Adapt the GSA Heuristic

We consider a distribution of random module lattices M of fixed determinant D such that, denoting $(\mathbf{b}_k, I_k)_k$ a pseudobasis of $M \subset K_{\mathbb{R}}^{\beta_K}$, and $(\mathbf{b}_k^*)_k$ its Gram-Schmidt orthogonalisation, generating the lattice M^* :

$$\forall k \in \llbracket 1; \beta_K - 1 \rrbracket, \mathbb{E}_M \{\ln(\det(\mathbf{b}_k^* I_k))\} = \ln(\alpha_K) + \mathbb{E}_M \{\ln \det(\mathbf{b}_{k-1}^* I_{k-1})\}.$$

Which yields, as $\det(\mathbf{b}_1^* I_1 + \dots + \mathbf{b}_{\beta_K}^* I_{\beta_K}) = \det(M) = D$:

$$\frac{\beta_K(\beta_K - 1)}{2} \ln(\alpha_K) = \beta_K \mathbb{E}_M \{\ln \det(\mathbf{s}I)\} - \ln D,$$

where $\mathbf{s}$ is a shortest vector of $\text{Module}(M^*)$ and I such that $\mathbf{s}I = \mathbf{s}K \cap M^*$.
Expressing $\mathbb{E}_M \{\ln \det(\mathbf{s}I)\}$ provides the terms composing α_K .

Expressing the Module-BKZ (mBKZ) Slope α_K

We have $\frac{\beta_K-1}{2} \ln(\alpha_K) = \mathbb{E}_M\{\ln \det(\mathbf{s}I)\} - \frac{\ln D}{\beta_K}$, where:

$$\mathbb{E}_M\{\ln \det(\mathbf{s}I)\} = \frac{1}{2}|\Delta_K| + \frac{1}{2}\mathbb{E}_M\{\ln \mathcal{N}(\bar{\mathbf{s}}^T \mathbf{s})\} + \mathbb{E}_M\{\ln \mathcal{N}(I)\}.$$

So:

$$\frac{\ln \alpha_K}{d} = \frac{2}{\beta_K - 1} (t_1 + t_2 + t_3 + t_4)$$

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where:

$$t_1 = \mathbb{E}_M \left\{ \ln \frac{\|\mathbf{s}\|}{D^{\frac{1}{d\beta_K}}} \right\}$$

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$$t_1 = \mathbb{E}_M \left\{ \ln \frac{\|\mathbf{s}\|}{D^{\frac{1}{d\beta_K}}} \right\} \quad t_2 = \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d}$$

Expressing the Module-BKZ (mBKZ) Slope α_K

We have $\frac{\beta_K - 1}{2} \ln(\alpha_K) = \mathbb{E}_M \{\ln \det(sI)\} - \frac{\ln D}{\beta_K}$, where:

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$$\begin{aligned} t_1 &= \mathbb{E}_M \left\{ \ln \frac{\|s\|}{D^{\frac{1}{d\beta_K}}} \right\} & t_2 &= \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d} \\ t_3 &= \mathbb{E}_M \left\{ \ln \frac{\sqrt{d} \mathcal{N}(\bar{s}^T s)^{\frac{1}{2d}}}{\|s\|} \right\} & t_4 &= \frac{\mathbb{E}_M\{\ln \mathcal{N}(I)\}}{d} \end{aligned}$$

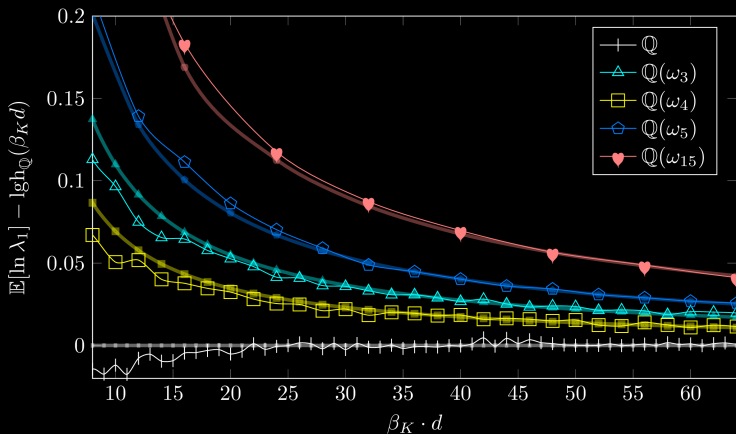
The Gaussian Heuristic Term $t_1 = \mathbb{E}_M \left\{ \ln \frac{\|s\|}{D^{\frac{1}{d\beta_K}}} \right\}$

The generalized Gaussian Heuristic for modules provides an estimation of t_1 .

For K a field with μ_K roots of unity, denoting

$\beta := d\beta_K$:

$$t_1 \approx \frac{1}{\beta} \ln \left(\frac{\mu_K}{2\text{Vol}(\mathcal{B}_\beta)} \right)$$



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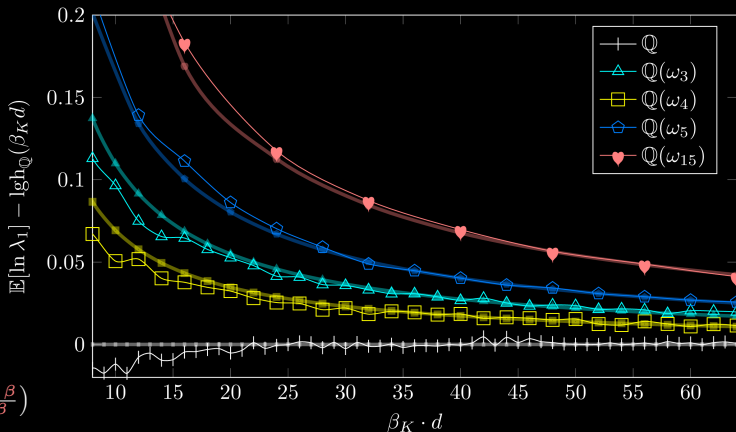
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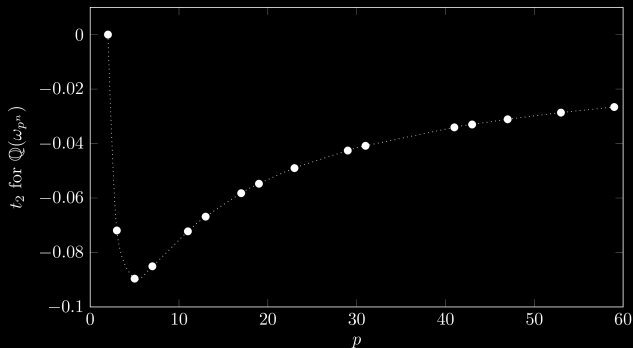
$$= \frac{\ln(\beta+2)}{2} - \frac{\ln(2\pi e)}{2} + \frac{\ln \beta}{2\beta} + o\left(\frac{\ln \beta}{\beta}\right)$$



The Discriminant Term $t_2 = \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d}$

Constant in β_K .

We calculate this term for cyclotomic fields $K = \mathbb{Q}(\omega_c)$.



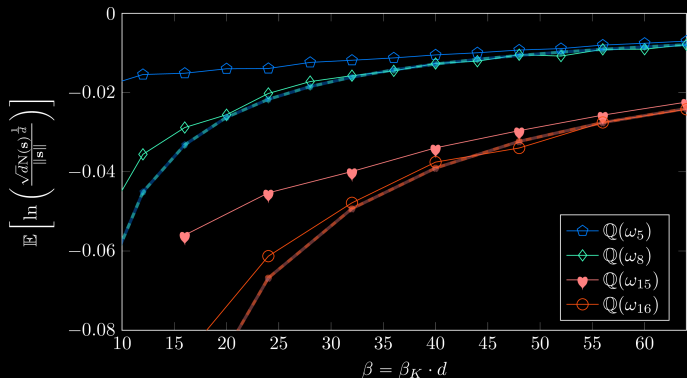
Then: $t_2 = \frac{1}{2} \sum_{\substack{p|c \\ p \text{ prime in } \mathbb{N}}} \left(\frac{p-2}{p-1} \ln(p) - \ln(p-1) \right).$

The Skewness Term $t_3 = \mathbb{E}_M \left\{ \ln \frac{\sqrt{d} \mathcal{N}(\bar{\mathbf{s}}^T \mathbf{s})^{\frac{1}{2d}}}{\|\mathbf{s}\|} \right\}$

Using the canonical embedding, the arithmetico-geometric inequality gives $t_3 \leq 0$.

Then, modelling the shortest vector as uniform of a sphere yields, denoting ψ the digamma function, $(d_{\mathbb{R}}, d_{\mathbb{C}})$ the signature of K , and $\beta = d\beta_K$:

$$t_3 \approx \frac{1}{2} \left(\frac{d_{\mathbb{R}} \psi\left(\frac{\beta_K}{2}\right) + 2d_{\mathbb{C}}(\psi(\beta_K) - \ln 2)}{d} + \ln d - \psi\left(\frac{\beta_K d}{2}\right) \right)$$

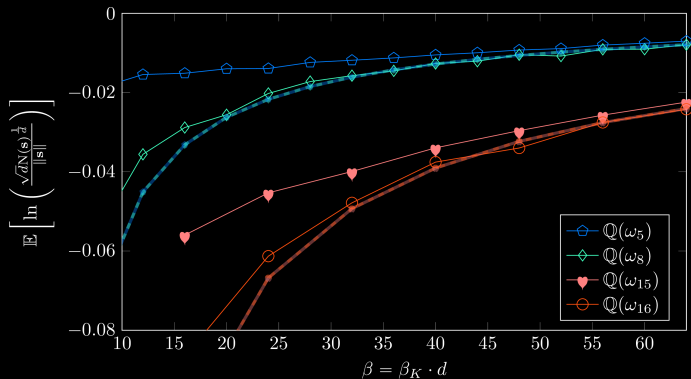


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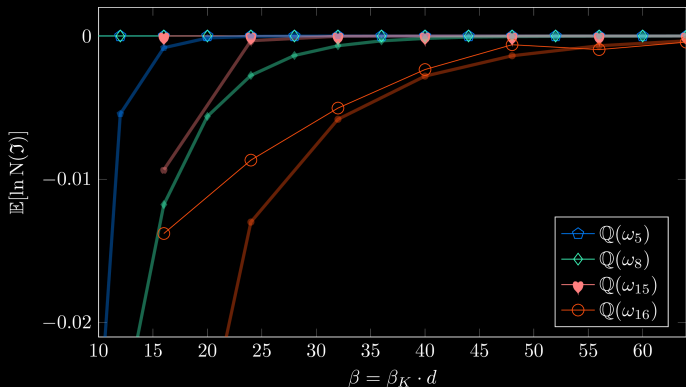


The Index Term $t_4 = \frac{\mathbb{E}_M\{\ln \mathcal{N}(I)\}}{d}$

The pseudobases are normalized to have $\mathcal{O}_K \subset I$, so: $t_4 \leq 0$.

Modelling M 's distribution to admit a K -basis B such that, for s a shortest vector, $s = Bx$ for x random large^a elements $\mathcal{O}_K$ yields, denoting $\frac{\zeta'_K}{\zeta_K}$ the logarithmic derivative of the zeta function of K :

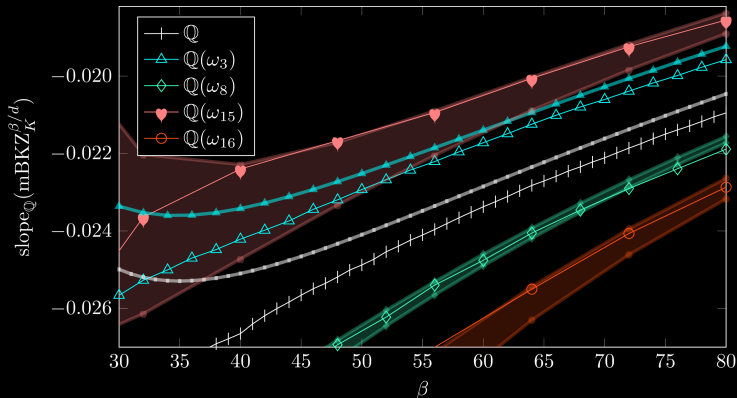
$$t_4 \gtrsim \frac{1}{d} \frac{\zeta'_K(\beta_K)}{\zeta_K(\beta_K)} \geq -\frac{1}{2^{\beta_K-1}} + o\left(\frac{1}{2^{\beta_K}}\right)$$



^asem: This gives an underestimation of the term

Comparing mBKZ and BKZ's Performances

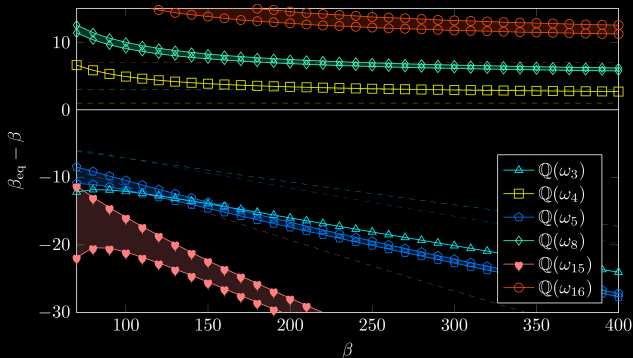
We implemented mBKZ and compare experimental slopes with our analysis bounding the $(t_i)_i$ terms. Implementation built on top of pfLLL and g6k.



Comparing mBKZ and BKZ's Performances

Denoting α the slope and β the block dimension in the unstructured case, $\frac{\ln \alpha_K}{d} = \alpha$, yields, denoting $\beta_{\text{eq}} = d\beta_K$:

$$\beta_{\text{eq}} = \beta + \log \left(\frac{|\Delta_K|}{d^d} \right) \frac{\beta}{d \ln \frac{\beta}{2\pi e}} + d - 1 + o(1)$$



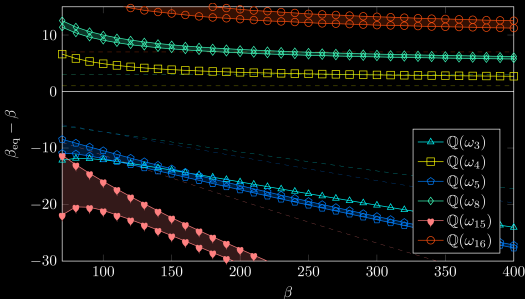
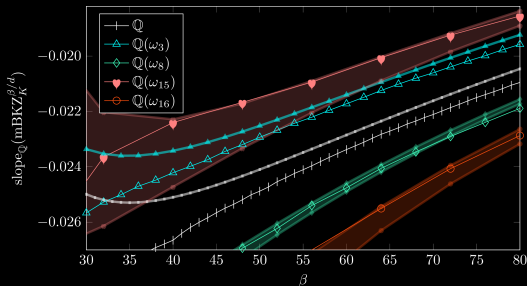
Next Open Questions

1. HKZ profile, tails and dimensions for free
2. Simulation
3. Lattice sieving using cyclotomic symmetries
4. Coefficient embeddings

Conclusion

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Thank you for your attention!
Do you have questions?

Bibliography

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